TECHNIQUE MASTERING EXERCISES

A - ABSOLUTE VALUE (Unit 1.1)

1. Write in the form $\begin{cases} \dots & \text{if } x \geq \dots \\ \dots & \text{if } x < \dots \end{cases}$.

a. $|x-2|$ **b.** $|3+6x|$ **c.** $|x^2-1|$ **d.** $|x^4+4|$

2. Solve, using the relevant theorems, and write your answers in interval notation (where possible).

a. $|x-2| = 0$ **b.** $|x-2| > 0$ **c.** $|x-2| < 0$ **d.** $|x-2| < 1$

e. $|2x| < 7$ **f.** $|2x-4| > 8$ **g.** $|x-3| \leq 6$ **h.** $|2x+5| \geq 6$

i. $|1-2x| = 3$ **j.** $|3-7x| \leq 6$ **k.** $|5-3x| = -3$ **l.** $|-5x-8| > 12$

m. $|x| = |-3|$ **n.** $|3x| = |2x+1|$ **o.** $\dfrac{x-1}{|2x|} \leq 0$ **p.** $\dfrac{|2x|}{x-1} \leq 0$

3. Sketch the graphs of the following pairs of functions.

a. $f(x) = x+4$ and $g(x) = |x+4|$ **b.** $f(x) = 1-x^2$ and $g(x) = |1-x^2|$

c. $f(x) = -x^3$ and $g(x) = |-x^3|$ **d.** $f(x) = 2^x$ and $g(x) = |2^x|$

B - INEQUALITIES

1. Solve and write your answers in interval notation.

a. $\dfrac{3-x}{-4} \geq$ **b.** $\dfrac{-4}{3-x} < 0$ **c.** $\dfrac{2x-6}{3x+9} \geq 0$

d. $\dfrac{x-1}{x} < 4$ **e.** $\dfrac{|-4x|}{3-x} < 0$ **f.** $\dfrac{2x-6}{|3x+9|} \geq 0$

2. Use graps to solve the inequlities and write your answers in interval notation.

a. $x^2+4 > 5$ **b.** $x^2-4 > 2-x$

♣**c.** $x^6 \geq x^4$ ♣**d.** $x^5 < x^7$

♣ - after you have done Unit 1.5

C - FUNCTIONS

1. Determine the domain of f.
Calculate the function value $f(a)$ of f at the given point a.

a. $f(x) = \dfrac{1}{\sqrt{x-1}}$, $a = 5$ **b.** $f(x) = \dfrac{3}{|x-4|}$, $a = 1$ **c.** $f(x) = x^4+x^3+1$, $a = 3$

d. $f(x) = \dfrac{1}{x^2+x+1}$, $a = 0$ ♣**e.** $f(x) = \dfrac{1}{\sin x}$, $a = \dfrac{5\pi}{2}$ ♦**f.** $f(x) = e^{\frac{1}{x}}$, $a = \ln 3$

♣ - after you have done Unit 1.4 ♦ - after you have done Units 1.7 and 1.9

a. $f(x) = \dfrac{1}{\sqrt{x-1}},\ a = 5$ **b.** $f(x) = \dfrac{3}{|x-4|},\ a = 1$ **c.** $f(x) = x^4 + x^3 + 1,\ a = 3$

d. $f(x) = \dfrac{1}{x^2 + x + 1},\ a = 0$ ♣**e.** $f(x) = \dfrac{1}{\sin x},\ a = \dfrac{5\pi}{2}$ ♦**f.** $f(x) = e^{\frac{1}{x}},\ a = \ln 3$

g. $f(x) = \dfrac{\sqrt{x}+1}{\sqrt{x}-1},\ a = 2$ ♣**h.** $f(x) = \tan x,\ a = -\dfrac{\pi}{4}$ **i.** $f(x) = \dfrac{|x|}{x},\ a = -1$

j. $f(x) = \dfrac{x^2 + 2x + 1}{x^2 - 1},\ a = 0$ ♦**k.** $f(x) = \dfrac{\ln x}{x-1},\ a = e$ ♦**l.** $f(x) = \ln(1 + e^x),\ a = 0$

♣ - after you have done Unit 1.4 ♦ - after you have done Units 1.7 and 1.9

2. For the given functions f and g, determine $f+g,\ f-g,\ fg,\ \dfrac{f}{g},\ f \circ g$ and $g \circ f$. .

a. $f(x) = 2x + 5,\quad g(x) = x^2$ **b.** $f(x) = x^2,\quad g(x) = \dfrac{1}{x-1}$

♦**c.** $f(x) = e^x,\quad g(x) = \ln x$ ♣**d.** $f(x) = \cos x,\quad g(x) = \dfrac{1}{x}$

e. $f(x) = \sqrt{x},\quad g(x) = x - 1$ **f.** $f(x) = \sqrt{x},\quad g(x) = 2^x$

♣ - after you have done Unit 1.4 ♦ - after you have done Units 1.7 and 1.9

D - TRIGONOMETRIC FUNCTIONS

1. Find, without using a calculator.

a. $\sin(\frac{7\pi}{4})$ **b.** $\operatorname{cosec}(-\frac{7\pi}{4})$ **c.** $\sin(35\pi)$

d. $\cos(-\frac{5\pi}{3})$ **e.** $\sec(\frac{5\pi}{3})$ **f.** $\sec(\frac{15\pi}{2})$

g. $\tan(\frac{7\pi}{6})$ **h.** $\cot(-\frac{7\pi}{6})$ **i.** $\cos(-\frac{15\pi}{3})$

2. Find, without using a calculator.

a. $\cos\theta, \tan\theta,\ \operatorname{cosec}\theta$ and $\cot\theta$ if $\sin\theta = \frac{7}{25},\ \theta \in (\frac{\pi}{2}, \pi)$

b. $\sin\theta, \tan\theta, \sec\theta$ and $\operatorname{cosec}\theta$ if $\cos\theta = -\frac{4}{5},\ \theta \in (\frac{\pi}{2}, \pi)$

c. $\sin\theta, \cos\theta,\ \cot\theta$ and $\sec\theta$ if $\tan\theta = \frac{5}{12},\ \theta \in (\pi, \frac{3\pi}{2})$

d. $\sin\theta, \tan\theta, \cot\theta$ and $\operatorname{cosec}\theta$ if $\cos\theta = -1,\ \theta \in (\frac{\pi}{2}, \frac{3\pi}{2})$

E - ALGEBRAIC OPERATIONS WITH EXPONENTIALS, LOGARITHMS, ETC

1. Express the following as a power of 2.

a. $\dfrac{2}{\sqrt{2}}$ **b.** $\sqrt[4]{8}$ **c.** 2.4^{2x}

d. $2^{x+y} \div 2^{x-y}$ **e.** $(2^x)^x$ **f.** $\dfrac{2^5 \cdot 2^{1/5} \cdot 2^{3/5}}{2^{-1/5}}$

2. Simplify.

a. $3a^4b^3 \cdot 2a^3b^{-2}$ **b.** $\dfrac{12x^{2n+3}}{4x^{n-1}}$ **c.** $(3x^2y^{-3}z^0)^{-4}$ **d.** $(3^{x+1})^{x+1}$

e. $(x^{1/y})^{1-y}$ **f.** $\sqrt{9x^2}$ **g.** $(4a^{-2}b^6)^{-\frac{1}{2}}$ **h.** $\left(\dfrac{\sqrt{x}}{\sqrt{x^{-5}}}\right)^{-\frac{1}{2}}$

i. $\dfrac{7^{-1}x^{1/4}}{3^{-2}y^{-1}} \div \dfrac{2(3^{-1/2})^2}{(49y)^0}$ **j.** $\dfrac{(2^{n-1})^{n-1}}{(2^{n+1})^{n+1}}$ **k.** $\left(\dfrac{16a^3}{81a^{-1}}\right)^{-\frac{3}{4}}$ **l.** $\dfrac{x^{8n-3}(xy^{1/2})^{-6n}x^3}{(xy)^{-4n}(x^{-1})^n}$

m. $\dfrac{12^n \times 8^{n-1} \times 3^{n+1}}{24 \times 6^{n-2} \times 2^n}$ **n.** $\sqrt[3]{a} \cdot \sqrt{a}$ **o.** $\dfrac{(3^n)^{1-n}}{6^{-1}\,3^{-n-1}} \div \dfrac{(3^{1-n})^{n+1}}{2^{-1}\,9^{-n-1}}$ **p.** $\dfrac{9^n - 3^{2n+1}}{(3^n)^2 - 3^{n+2}\,3^n}$

q. $\dfrac{5^x - 3^2 5^{x-2}}{5^{x+1} + 3.5^x}$ **r.** $(a^{1/2} - b^{3/2})^2$ **s.** $\dfrac{a^{-1}}{x^{-1} - a^{-1}}$ **t.** $\left(\dfrac{(3y)^{-2}}{3y^{-2}}\right)^{\frac{2}{3}}$

3. Express each logarithm as the sum or difference of simpler logarithms.

a. $\log\left(\dfrac{xy}{z}\right)$ **b.** $\log\sqrt{\dfrac{x}{y}}$ **c.** $\log(x^2y^3)$ **d.** $\log\left(\dfrac{1}{x^2y}\right)$

e. $\ln\left(\dfrac{xy^2}{x+y}\right)$ **f.** $\ln\sqrt[3]{x^2+1}$ **g.** $\ln\sqrt[3]{x^2y}$ **h.** $\ln\sqrt{\dfrac{x^3}{y}}$

i. $\log\left(\dfrac{x}{yz}\right)$ **j.** $\ln\left(\dfrac{(x+y)^3}{xy^2}\right)$

4. Express each statement as a single logarithm with coefficient 1.

a. $3\ln x$ **b.** $3\ln x - \frac{1}{2}\ln y$ **c.** $\frac{1}{2}(\ln x - 3\ln y)$

d. $2\ln x + \ln(x+y)$ **e.** $2\ln x - 3\ln y + \ln z$ **f.** $\ln(y-2) + \ln y - 2\ln x$

g. $\frac{1}{2}[\ln x - 5(\ln y - 3\ln z)]$ **h.** $\frac{1}{2}[(\log x - 5\log y) - 3\log z]$ **i.** $\log 5 + 3\log x - 2\log 4 - \log y$

j. $3\log(xy) - 2\log x - \log y$

5. If $\log_a 2 = x$ and $\log_a 3 = y$ express each of the following in terms of x and y.

a. $\log_a 27$ **b.** $\log_a \dfrac{2}{9}$ **c.** $\log_a \sqrt[3]{6}$ **d.** $\log_a \sqrt{12a}$

6. If $f(x) = 2^x$ determine the equations of g, h and k if

a. g is symmetrical to f about the x –axis.
b. h is symmetrical to f about the y –axis.
c. k is symmetrical to f about the line $y = x$.

7. If $log_a b = 3$, find $log_b a$.

8. Show that if $y = \dfrac{r}{1 + ce^{-at}}$ then $t = \dfrac{1}{a}\ln\dfrac{cy}{r-y}$.

9. If $f(x) = log_b x$, find $f(\frac{1}{b})$ and $f(\sqrt{b})$.

10. If $log_b(\frac{1}{10}) = -\dfrac{1}{2}$, find b.

11. Simplify.

a. $4(e^x)^4 + (x^e)^4$ **b.** $\frac{1}{2}e^{2x} \cdot 4e^{3/x}$ **c.** $\frac{14e^{3x}}{7e^x}$

d. $\sqrt{e^x} \cdot \sqrt[3]{e^{-3x}}$ **e.** $\ln(e^{3x})$ **f.** $e^{\ln 7x}$

g. $\ln(9e^x \cdot 10e^{2x})$ **h.** $e^{\ln(\frac{x^2}{y})}$ **i.** $3\ln x + \ln \frac{1}{x}$

j. $\ln e^2 + e^{-\ln x}$

12. Solve for x without using a calculator.

a. $\ln e^{4x} = 12$ **b.** $e^{4x} = 12$ **c.** $\ln 4x = 12$

d. $e^{2\ln 3} = x$ **e.** $\ln x = 3\ln 2 - 2\ln 3$ **f.** $\frac{1}{2}\ln x = 1 - \ln 2$

F - GRAPHS

1. Sketch the graphs of the following functions.

a. $f(x) = 2x + 3 \quad x \in [-4, 3]$ **b.** $f(x) = |2x - 1|$

c. $f(x) = |x| + 1$ **d.** $f(x) = \begin{cases} x^2, & x \leq 1 \\ -x + 2, & x > 1 \end{cases}$

e. $f(x) = x^2 - x - 6$ **f.** $f(x) = \sqrt{9 - x^2} \quad x \in [-3, 3]$

g. $.f(x) = 2\sin x + 1 \quad x \in [-2\pi, \pi]$ **h.** $f(x) = \cos\frac{x}{2} \quad x \in [0, 2\pi]$

i. $f(x) = \tan 2x \quad x \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

2. Determine the domain of the function and sketch the function (After you have done Units 1.6 to 1.9)

a. $f(x) = \ln x$ **b.** $g(x) = \ln(-x)$ **c.** $h(x) = \ln|x|$

d. $l(x) = |\ln x|$ **e.** $m(x) = \ln(1 + x)$ **f.** $m(x) = 1 + \ln x$

3. Determine the domain of the function and sketch the function (After you have done Units 1.6 to1.9)

a. $f(x) = e^x$ **b.** $g(x) = e^{-x}$ **c.** $h(x) = e^{|x|}$

d. $l(x) = |e^x|$ **e.** $m(x) = e^{1+x}$ **f.** $n(x) = 1 + e^x$

G - GRAPHS (After you have done Units 1.7, 1.9, 1.10 and 3.7)

Sketch the graphs of the following pairs of functionson the same set of axes.

1. $f(x) = \sinh x$ and $g(x) = 2\sinh x + 1$

2. $f(x) = \cosh x$ and $g(x) = \cosh\frac{x}{2}$

3. $f(x) = \tanh x$ and $g(x) = \tanh 2x$

4. $f(x) = \ln\ x$ and $g(x) = \ln 2x$

5. $f(x) = \ln\ x$ and $g(x) = \ln(x + 2)$

6. $f(x) = \ln x$ and $g(x) = 2\ln x$

7. $f(x) = e^x$ and $g(x) = e^{2x}$

8. $f(x) = e^x$ and $g(x) = \ln x$

9. $f(x) = \sin x$ and $g(x) = \arcsin x$

10. $f(x) = \cos x$ and $g(x) = \arccos x$

11. $f(x) = \tan x$ and $g(x) = \arctan x$

12. $f(x) = \arcsin x$ and $g(x) = \arcsin(x+1)$

H - LIMITS

Determine the following limits whenever they exist.

1. $\lim\limits_{x\to 0} (x + \frac{1}{x})$
2. $\lim\limits_{x\to 1} \dfrac{x^2-1}{x^2-2x+1}$
3. $\lim\limits_{x\to 3} \dfrac{x^2-9}{x-3}$
4. $\lim\limits_{t\to 0} \dfrac{t^2-t}{t}$
5. $\lim\limits_{x\to 2} \dfrac{\pi}{2}$
6. $\lim\limits_{x\to 0} |x|$
7. $\lim\limits_{x\to e} \ln x^2$
8. $\lim\limits_{x\to \frac{\pi}{2}} \tan x$
9. $\lim\limits_{x\to \frac{\pi}{2}} (1-\sin x)$
10. $\lim\limits_{u\to 2} (u^3+4u+1)$
11. $\lim\limits_{x\to \infty} (x + \frac{1}{x})$
12. $\lim\limits_{x\to \infty} \dfrac{x^2+x+1}{x^3+2x^2+1}$
13. $\lim\limits_{x\to 0^+} \ln x$
14. $\lim\limits_{x\to 3^-} \dfrac{-1}{\sqrt{3-y}}$
15. $\lim\limits_{x\to 1^+} \dfrac{x^2-3x+2}{x^2-2x+1}$

I - DIFFERENTIATION

Differentiate the following functions.

1. $f(x) = 2x^4 - 3x^2 + 5x + 2$
2. $f(x) = x^{-1/2} + x^{1/2}$
3. $f(x) = 3x^2 + \frac{1}{3}x^{-2} + x$
4. $f(x) = x^2 - 3x^{7/3}$
5. $f(t) = t + 1 + \dfrac{1}{t^2}$
6. $f(x) = \sqrt{x} - \dfrac{1}{\sqrt{x}}$
7. $f(x) = \dfrac{12}{x} - \dfrac{4}{x^3} + \dfrac{1}{x^4}$
8. $f(x) = \sqrt[3]{x} + \dfrac{1}{x^3}$
9. $f(x) = x^4 - 8x^3 + 2x^2 - x + 1$
10. $f(y) = 7y^3 - 5y^2 + 3y - 17$
11. $f(x) = 9x^{-3} + 2x^{-2} - 14$
12. $f(x) = -2x^4 + x^{-2} - 3x^{3/4}$
13. $f(t) = 12t^4 + 3t^3 + 5t^{-2} - 4$
14. $f(x) = 4x^{-2} - 7\sqrt{x} + 8x^3 + 5$
15. $f(x) = 3x^3 + 2x^2 - x + 1$
16. $f(x) = 4x^3 - 7x^2 + 8x - 6$
17. $f(x) = x^3 - 3x - 2x^{-4}$
18. $f(x) = x + 3 + \dfrac{4}{x}$
19. $(u) = -3u^{-3} + 4u^2 + \dfrac{1}{u^2}$
20. $f(x) = x\sqrt{x} + \dfrac{1}{x^2\sqrt{x}}$

21. $f(x) = (x^3 + 7x^2 + 5)^5$ **22.** $f(x) = \sqrt{x^4 + x^2 + 2}$ **23.** $f(x) = (\sqrt{x+1})(x^2 + 1)$

24. $f(t) = \sqrt{t^2 + 2t + 3}$ **25.** $f(x) = x^2(x^3 - 1)$ **26.** $f(x) = (x^3 - 3x)^4$

27. $f(x) = x\sqrt{1 - x^2}$ **28.** $f(x) = (2x - 4)(3x^2 + 2)$ **29.** $f(x) = (2x - 3)^3(4x + 2)^2$

30. $f(x) = (7x + 3)^2(3x^2 - 14x + 5)$

31. $f(y) = (8y^3 - 2)(3y^2 - 5y + 10)^2$ **32.** $f(x) = (2x^3 - 3)^{2/3}$ **33.** $f(x) = (3x^2 - 2x + 1)^{1/2}$

34. $f(x) = (2x - 3)(3x + 4)$ **35.** $f(x) = (2x^3 - 1)(x^4 + x)$ **36.** $f(x) = (\frac{1}{x} + 3)(x^2 - 5)$

37. $f(x) = (2x + 1)^2(x^2 + 2)^3$ **38.** $f(x) = \sqrt{3x^2 + 1}$ **39.** $f(t) = (2t - 5)^3$

40. $f(x) = (6x - 5)^{-3}$

41. $f(x) = \dfrac{x^2 + x + 1}{x^2 + 1}$ **42.** $f(x) = \dfrac{2x + 3}{3x + 2}$ **43.** $f(u) = \dfrac{2u + 1}{3u - 5}$

44. $f(x) = \dfrac{x^2 + 5x - 1}{x^2}$ **45.** $f(x) = \dfrac{(x + 1)(x + 2)}{(x - 1)(x - 2)}$ **46.** $f(x) = \sqrt{\dfrac{x^2 + 1}{x^2 + 4}}$

47. $f(t) = \dfrac{t + t^3}{\sqrt{t}}$ **48.** $f(x) = \left(\dfrac{x + 1}{x - 1}\right)^2$ **49.** $f(x) = \dfrac{10}{\sqrt{x} + 4}$

50. $f(x) = \dfrac{8}{4 + x^2}$

51. $f(y) = \dfrac{4y}{y^2 + 1}$ **52.** $f(x) = \dfrac{3}{\sqrt{2x + 1}}$ **53.** $f(x) = \dfrac{3x^2 - 2x + 3}{4x^2 - 5}$

54. $f(x) = \dfrac{(2x - 4)(3x + 5)}{2x^2 + 7}$ **55.** $f(x) = \dfrac{2x - 3}{x^2 + 2x}$ **56.** $f(t) = \dfrac{4 - 2t + 3t^2}{t^2 + 2}$

57. $f(x) = 2x^3 + \dfrac{2 - x}{x^3}$ **58.** $f(x) = \dfrac{x - 1}{x + 1}$ **59.** $f(x) = \dfrac{x^2}{x^2 + 1}$

60. $f(t) = \dfrac{1}{t^4 - 2t + 1}$

61. $f(x) = \sin(x^2 + 2)$ **62.** $f(x) = \sin(3x)$ **63.** $f(y) = \cos(3y)$

64. $f(x) = \cos(x^4 + 7)$ **65.** $f(x) = \tan(\sin x)$ **66.** $f(t) = \tan(t^2 + 5)$

67. $f(x) = \sin(3x + 2)$ **68.** $f(x) = 2\sin x - \tan x$ **69.** $f(x) = 1 + x - \cos x$

70. $f(x) = \dfrac{\sin x}{x}$

71. $f(x) = \tan(2x - x^3)$ **72.** $f(\theta) = \sin(3\theta^2 - 2\theta + 1)$ **73.** $f(x) = 3\sin(x^2) + 2\sin(x) - \sin(3)$

74. $f(x) = 3\sin x - 2\cos x$ **75.** $f(x) = -\cos(\pi x - 1)$ **76.** $f(x) = \cos(3x) - \tan(3x)$

77. $f(x) = x^2 \sin x$ **78.** $f(x) = \cos(\tan x)$ **79.** $f(x) = \sin(2x + 3)$

80. $f(t) = \sin(2\pi t)$

81. $f(x) = \sin\left(\frac{1}{\sqrt{x}}\right)$ **82.** $f(x) = \sin\left(\frac{1}{x}\right)$ **83.** $f(x) = \cos(-x)$

84. $f(x) = \tan\pi(\frac{1}{2} - x)$ **85.** $f(x) = \tan(x^2 + 1)$ **86.** $f(t) = \tan(5t)$

87. $f(x) = x\cos x$ **88.** $f(x) = \frac{\sin x}{1 - x}$ **89.** $f(x) = 4\sin^2(3x)$

90. $f(x) = \frac{\cos x}{\sin x}$

91. $f(x) = (x^2 + 3)\sin x$ **92.** $f(x) = \sqrt{\sin x}$ **93.** $f(x) = \cos^2(x^3)$

94. $f(x) = x\cos(5x)$ **95.** $f(y) = \sin(2y)\cos(3y)$ **96.** $f(x) = \sin^2(3x) + \cos^2(5x)$

97. $f(x) = x^2\tan\left(\frac{1}{x}\right)$ **98.** $f(x) = (\sin x - x\cos x)^{-1}$ **99.** $f(x) = \frac{\sin x + \cos x}{\tan x}$

100. $f(x) = \sin^2\left(x + \frac{1}{x}\right) + \cos^2\left(x + \frac{1}{x}\right)$

101. $f(x) = \frac{x^2}{\tan x}$ **102.** $f(x) = \sqrt{\cos(2x)}$ **103.** $f(x) = \sin^2(x - 3)$

104. $f(t) = \cos^3(t^2 - t)$ **105.** $f(x) = \sin x\cos^2 x$ **106.** $f(x) = \sin^{1/3}(2x)$

107. $f(x) = 4\sin^7(2 - 4x)$ **108.** $f(x) = 2\sin x + 3\sin^2 x$ **109.** $f(u) = 3\cos^3 u + 4\cos^2 u - 6$

110. $f(x) = 5\cos^2(x + 2) + 3\cos(x + 2) - 5$

111. $f(x) = 5x^2 - 3\tan^2 x + \sin x$ **112.** $f(x) = \frac{\tan x}{1 - \tan x}$ **113.** $f(x) = (\sin(x + 1))^{3/2}$

114. $f(t) = \sin^2 t + \cos^2 t$ **115.** $f(x) = \sin\sqrt{x} + \sqrt{\sin x}$ **116.** $f(x) = (\sin x)(\sin x + \cos x)$

117. $f(x) = 2\sin x\cos x$ **118.** $f(u) = \frac{1}{\sin u}$ **119.** $f(x) = \cos^2 x\sin x$

120. $f(x) = \sqrt{1 - \sin^2 x}$

121. $f(x) = x + \tan^2 x$ **122.** $f(x) = \frac{x}{\cos x}$ **123.** $f(x) = \tan x\cos x$

124. $f(t) = \sin t\cos t$ **125.** $f(x) = e^{x^2}$ **126.** $f(x) = e^{5x}$

127. $f(x) = (x^2 + 3x)e^x$ **128.** $f(x) = xe^x - e^{-x}$ **129.** $f(x) = e^{x^2} \bullet e^{x+1}$

130. $f(x) = \frac{e^{x^2}}{e^{x-1}}$

131. $f(u) = e^{-u}$ **132.** $f(x) = e^{\frac{2x}{3}}$ **133.** $f(x) = e^{\sqrt{x}}$

134. $f(x) = e^{3x} + 2e^{2x} - 3e^x + 7$ **135.** $f(x) = e^{x^2-2}$ **136.** $f(x) = \frac{1 + e^{2x}}{2 - e^{2x}}$

137. $f(x) = e^{3x-1} - 4e^{-x}$ **138.** $f(t) = \cos(e^t)$ **139.** $f(x) = 3e^{2x} - 4e^x + 1$

140. $f(x) = e^{3\cos(2x)}$

141. $f(x) = e^{-2x} + 4e^{-3x} + 7$ **142.** $f(x) = e^{2x+1}$ **143.** $f(x) = \frac{1}{2}e^{2x}$

144. $f(x) = e^{\sin x}$ **145.** $f(x) = e^{2x}$ **146.** $f(t) = 2te^{t}$

147. $f(x) = \dfrac{1}{1 - e^{-x}}$ **148.** $f(x) = \dfrac{e^{-x}}{x}$ **149.** $f(x) = x^2 e^{-x}$

150. $f(x) = e^{-\frac{1}{x^2}}$

151. $f(x) = e^{\sqrt{x^2+1}}$ **152.** $f(x) = \dfrac{e^x - e^{-x}}{2}$ **153.** $f(x) = e^{2x}\cos(3x)$

154. $f(x) = e^{\cos(4x)}$ **155.** $f(x) = x^2 \cdot 2^x$ **156.** $f(x) = 3^{5x}$

157. $f(x) = x^4 + 4^x$ **158.** $f(x) = 9^{-x}$ **159.** $f(x) = \tan(5^x)$

160. $f(x) = 3^{4x+1} + 2^{4x+2}$

161. $f(y) = 3^{y^2+1}$ **162.** $f(x) = 2^x$ **163.** $f(x) = 2^{-x}$

164. $f(t) = (\frac{1}{2})^t$ **165.** $f(x) = e^x \ln x$ **166.** $f(x) = \ln(\sin x)$

167. $f(x) = \dfrac{1}{\ln x}$ **168.** $f(x) = \ln(3xe^{-x})$ **169.** $f(x) = \ln\left(\dfrac{x-1}{x^2+1}\right)$

170. $f(x) = \ln\left(\frac{e^x}{1+e^x}\right)$

171. $f(x) = \ln(e^{\sin 2x})$ **172.** $f(x) = \ln\sqrt{\dfrac{x}{x^2+1}}$ **173.** $f(u) = \ln(u^2)$

174. $f(x) = \ln\left(\dfrac{10}{x}\right)$ **175.** $f(x) = \ln(10^x)$ **176.** $f(x) = \ln(3x) + 4\ln x$

177. $f(x) = x^2\ln(2x)$ **178.** $f(x) = \ln(x^{-1})$ **179.** $f(x) = x\ln x$

180. $f(x) = \ln\frac{1}{x}$

181. $f(y) = (\ln y)^3$ **182.** $f(x) = x\ln(\sqrt{x})$ **183.** $f(x) = \ln(7x)$

184. $f(x) = (\ln x)^{1/2}$

J - DIFFERENTIATION

Differentiate the following functions.

1. $f(x) = \sinh(x^2 + 4x)$ **2.** $f(x) = \sinh(7x)$ **3.** $f(x) = \cosh(x^4 + 5x + 7)$

4. $f(t) = \tanh(\sinh 2t)$ **5.** $f(x) = \tanh(x^2 + 5)$ **6.** $f(x) = \dfrac{\sinh x}{x}$

7. $f(y) = \tanh(2y - y^3)$ **8.** $f(x) = 3\sinh(x^2) + 2\cosh(x) - \tanh(3)$

9. $f(x) = 3\sinh 4x - 2\cosh 5x$ **10.** $f(x) = \cosh(\tanh x)$

11. $f(x) = \cosh\left(\dfrac{1}{\sqrt{x}}\right)$ **12.** $f(x) = \tanh\left(\dfrac{1}{x}\right)$ **13.** $f(t) = \sinh(-t)$

14. $f(x) = \dfrac{\cosh x}{1+x}$ **15.** $f(x) = 4\cosh^2(3x)$ **16.** $f(x) = \sinh^2(x^4)$

17. $f(x) = \ln(e^{\cosh 2x})$ **18.** $f(\theta) = \arcsin(2\theta)$ **19.** $f(x) = \arccos(x^2)$

20. $f(x) = x(\arcsin x)$

21. $f(x) = \arcsin\left(\frac{2}{x}\right)$ **22.** $f(y) = \arcsin(2y - 3)$ **23.** $f(x) = 2x(\arctan x)$

24. $f(x) = \arctan(5x)$ **25.** $f(t) = \arcsin(\sin t)$ **26.** $f(x) = \arctan(3x - 4)$

27. $f(x) = \arccos\left(\frac{x}{4}\right)$

K - IMPLICIT DIFFERENTIATION

The following equations define y implicitely as a function of x.

Determine $\frac{dy}{dx}$ in terms of x and y in each case.

1. $x^2 + y^2 = 100$ **2.** $x^3 - y^3 = 6xy$ **3.** $x^2y + 3xy^3 - x = 3$ **4.** $x^3y^2 - 5x^2y + x = 1$

5. $x^2 = \frac{x+y}{x-y}$ **6.** $\sqrt{xy} + 1 = y$ **7.** $(x^2 + 3y^2)^{35} = x$ **8.** $\cos xy = y$

9. $\sin(x^2y^2) = x$ **10.** $\tan^3(xy^2 + y) = x$ **11.** $\sqrt{3 + \tan xy} - 2 = 0$

L - ANTIDERIVATIVES

Find the most general form of the function f satisfying the following.

1. $f'(x) = x^8$ **2.** $f'(x) = \frac{1}{x^6}$ **3.** $f'(x) = x^{5/7}$ **4.** $f'(t) = \sqrt[3]{t^2}$

5. $f'(x) = \frac{4}{\sqrt{x}}$ **6.** $f'(x) = \frac{1}{2x^3}$ **7.** $f'(x) = x^3\sqrt{x}$ **8.** $f'(x) = (x^3 - 2x + 7)$

9. $f'(x) = (x^{-3} + \sqrt{x} - 3x^{1/4} + x^2)$

10. $f'(u) = (u^{2/3} - 4u^{1/5} + 4)$ **11.** $f'(x) = \left(\frac{7}{x^{3/4}} - \sqrt[3]{x} + 4\sqrt{x}\right)$

12. $f'(x) = x(1 + x^3)$ **13.** $f'(t) = (1 + t^2)(2 - t)$ **14.** $f'(x) = x^{1/3}(2 - x)^2$

15. $f'(x) = \left[\frac{1}{x^2} - \cos x\right]$ **16.** $f'(\theta) = [4\sin\theta + 2\cos\theta]$

17. $f'(y) = e^y$ **18.** $f'(x) = xe^{x^2}$ **19.** $f'(x) = \frac{2}{x}$

M - DIFFERENTIAL EQUATIONS

Find the solution of each of the following differential equations with initial values (This means you have to find the function y.).

1. $\frac{dy}{dx} = \frac{x}{2},\ y\left(\frac{1}{2}\right) = -1$ **2.** $\frac{dy}{dx} = -\frac{3}{2}x^2,\ y(-1) = -\frac{1}{2}$

3. $\frac{dy}{dt} = \sin t,\ y\left(\frac{\pi}{2}\right) = 3$ **4.** $\frac{dy}{dx} = e^x,\ y(0) = 4$

5. $\frac{dy}{dx} = \frac{1}{x},\ y(1) = 2$ **6.** $\frac{d^2y}{dx^2} = 0,\ \frac{dy(2)}{dx} = 1,\ y(2) = 2$

7. $\frac{d^2y}{dx^2} = \cos x,\ \frac{dy(0)}{dx} = 1,\ y(0) = 2$ **8.** $\frac{d^2y}{dt^2} = e^t,\ \frac{dy(1)}{dt} = e,\ y(1) = -4e$

N - INTEGRATION BY INSPECTION

Find the following integrals:

1. $\int e^{5x}\,dx$ **2.** $\int x^8 dx$ **3.** $\int \frac{1}{x^6} dx$

4. $\int t^{5/7} dt$ **5.** $\int \sqrt[3]{x^2}\,dx$ **6.** $\int \frac{4}{\sqrt{x}} dx$

7. $\int \frac{1}{2\theta^3} d\theta$ **8.** $\int x^3\sqrt{x}\,dx$ **9.** $\int (x^3 - 2x + 7)dx$

10. $\int (x^{-3} + \sqrt{x} - 3x^{1/4} + x^2)dx$

11. $\int (y^{2/3} - 4y^{1/5} + 4)dy$ **12.** $\int\left(\frac{7}{x^{3/4}} - \sqrt[3]{x} + 4\sqrt{x}\right)dx$ **13.** $\int x(1 + x^3)dx$

14. $\int(1 + u^2)(2 - u)du$ **15.** $\int x^{1/3}(2 - x)^2 dx$ **16.** $\int\left[\frac{1}{t^2} - \cos t\right]dt$

17. $\int[3\sin x - 4\cos x]dx$ **18.** $\int e^{-x}dx$ **19.** $\int 2xe^{x^2}dx$

20. $\int \frac{1}{2x} dx$

21. $\int e^{3t}\,dt$ **22.** $\int \frac{3}{x}\,dx$ **23.** $\int \frac{1}{2x + 1}\,dx$

24. $\int \sinh\,4x\,dx$ **25.** $\int \sinh\,(4x + 6)\,dx$ **26.** $\int \cosh\,(2x + 3)\,dx$

27. $\int \tanh\,3x\,dx$ **28.** $\int \frac{u^2}{u^3 - 4} du$ **29.** $\int \frac{5u^4}{u^5 + 1} du$

30. $\int \frac{t + 1}{t} dt$

31. $\int \frac{x^3}{x^2 + 1} dx$ **32.** $\int x^2(2 - x^3)^4\,dx$ **33.** $\int xe^{2x^2}\,dx$

34. $\int \sin\theta(\cos\theta - 3)^3\,d\theta$ **35.** $\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,dx$ **36.** $\int \frac{\sin\theta}{1 + \cos\theta}\,d\theta$

37. $\int \tan\theta\,d\theta$ **38.** $\int \frac{1}{x\,\ln\,x}\,dx$ **39.** $\int \frac{\ln\,y}{y}\,dy$

40. $\int(x + 1)\sqrt{x^2 + 2x}\,dx$